- When the interest is compounded annually, $A = P\left(1+\frac{r}{100}\right)^{n}$.
- When the interest is compounded half yearly, $A = P\left(1+\frac{r}{200}\right)^{2n}$.
- When the interest is compounded quarterly, $A = P\left(1+\frac{r}{400}\right)^{4n}$.
- When the interest is compounded annually but rate of interest differs year by year, $A = P\left(1+\frac{a}{100}\right)\left(1+\frac{b}{100}\right)\left(1+\frac{c}{100}\right)\ldots$ where $a$, $b$ and $c$ are interest rates for the I, II and III years respectively.
- When interest is compounded annually but time period is in fraction say $a\frac{b}{c}$ years, $A = P\left(1+\frac{r}{100}\right)^{a}\left(1+\frac{\frac{b}{c}\times r}{100}\right)$.
- C.I = A–P (Amount – Principal).
- The simple interest and the compound interest remains the same for the first year or the first conversion period.
- For 2 years, the difference in C.I and S.I is $C.I - S.I = P\left(\frac{r}{100}\right)^{2}$.
- For 3 years, the difference in C.I and S.I is $C.I - S.I = P\left(\frac{r}{100}\right)^{2}\left(3+\frac{r}{100}\right)$.
- $x$ and $y$ are said to vary directly if $\boldsymbol{y=kx}$ always, where $k$ is called the proportionality constant and $k > 0$ assuming that $y$ depends on $x$ and so $\boldsymbol{k=\frac{y}{x}}$.
- $x$ and $y$ are said to vary inversely, if $\boldsymbol{xy = k}$ always, where $k$ is called the proportionality constant and $k > 0$.
- There will be problems which involve a chain of two or more variations in them. This is called as compound variation.
- By finding the proportion, we can use the fact that the product of the extremes is equal to the product of the means to find the unknown ($x$) in the problem.
- By using the formula $\frac{P_1 \times D_1 \times H_1}{W_1} = \frac{P_2 \times D_2 \times H_2}{W_2}$, we can find the unknown ($x$).
- We can also find the unknown ($x$) by Multiplicative Factor Method.
- If two persons X and Y can complete some work individually in $\boldsymbol{a}$ and $\boldsymbol{b}$ days, their one day's work will be $\frac{\boldsymbol{1}}{\boldsymbol{a}}$ and $\frac{\boldsymbol{1}}{\boldsymbol{b}}$ respectively and X and Y together can complete the work in $\frac{\boldsymbol{ab}}{\boldsymbol{a+b}}$ days.



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